Tomas Ulises Traini

Software Engineer, Clinical Research Systems

Rosario, Argentina. Argentine and Spanish citizen, EU passport. Open to remote or relocation.
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SUMMARY

Full-stack engineer, about five years, mostly .NET and React, now Python and FastAPI. At CLINIGMA in Denmark I am the only engineer on ClinigmaScriber, a system that takes recorded patient interviews through transcription, translation and qualitative coding for clinical trials. Before that, React and SAP-backed microservices at John Deere, and .NET modernization and security work at PwC. English C2, German B1 and improving.

SKILLS

Languages: Python, TypeScript, C#, JavaScript, SQL

Backend: FastAPI, SQLAlchemy, Pydantic, .NET and .NET Core, NestJS, Node.js, REST, GraphQL, WebSockets

Frontend: React, TypeScript, Zustand, TanStack Query, Redux, Vite, Tailwind, Angular

Data and cloud: PostgreSQL, SQL Server, DynamoDB, Azure (App Service, Blob, OpenAI, Translator, Speech), AWS SQS, Terraform, Docker, Azure DevOps

Practice: hexagonal architecture, microservices, pytest, vitest, Cypress, Playwright, AI agent workflows, WCAG 2.1, Checkmarx

Clinical domain: ICH-GCP, GxP, 21 CFR Part 11, audit trails, e-signature workflows, qualitative coding, codebooks, QDPX (REFI-QDA), speech recognition, speaker diarization

EXPERIENCE

Technical Officer, Software Engineer (Consultant)

CLINIGMA ApS. Denmark, remote. Oct 2025 - Present

CLINIGMA runs in-trial patient interviews and produces qualitative patient-experience data for pharmaceutical research and regulatory submissions, under ICH-GCP and ISO 9001.

- Sole engineer on ClinigmaScriber, which turns recorded patient interviews into coded qualitative data. I chose the architecture, the stack and the coding standards, with the CTO reviewing. Python 3.11 and FastAPI in a hexagonal layout, React and TypeScript on the client, PostgreSQL 16, running on Azure. First internal pilot starts September 2026.
- Three stages, transcription then translation then coding, each one locked by a trial manager's signature. Every interview is its own embedded git repository, addressed sentence by sentence, with an append-only edit log behind it. Coding is text-frozen, so coders only add annotations against a project codebook; QDPX interop, the REFI-QDA format that ATLAS.ti and NVivo read, is the next piece of work. Built against GCP/GxP and 21 CFR Part 11 expectations, though it is not a validated system and I do not present it as one.
- Speech pipeline runs on Azure Fast Transcription, with word-level timestamps, diarization and language detection. I wrote a local CLI that scores Whisper, Qwen3 and pyannote on the same audio for word error rate and timing drift, which is how the backend got chosen.
- Roughly 4,500 backend and 2,800 front-end tests, plus Cypress against a live stack. I set the CI gates the repository runs on: a 500-line cap per file, layer import rules, locale parity, and a rule that every bug gets written into a ledger with the test that catches it again.
- Built a schema-driven system for compliance documents, then argued for killing it. The engineering held up. It asked users to decide whether a form was reusable or single use, and most of them work in Word, so it was the wrong thing to build.
- Ran agent experiments with my manager. A GitHub ticket goes into a Teams thread, then through refiner, architect and per-stack implementation agents, then QA and a pull request. Two users, both of us. That work fed into a large jQuery to React rewrite of the company portal, which he led and built.

Senior Software Developer

Sistran Consultores. Remote. Aug 2025 - Feb 2026

- React and .NET work on insurance policy workflows, inside an existing enterprise codebase and delivery process.

Ssr. Software Engineer

John Deere. Remote. Sep 2023 - Jun 2025

- Main React and TypeScript engineer on the tractor parts management apps, in a Node.js microservices setup with Terraform and CI/CD. Worked through a large front-end refactor: reducer-based state instead of prop drilling, a proper type convention instead of `any` and over-generic types, consistent naming.
- Added GraphQL queries and pagination for catalogs where one SAP query could return around 100,000 rows. Rendering that naively killed the page, so performance set the design.
- Owned the features that crossed into SAP. Connected the parts service, coordinated the SAP developer so their events landed alongside ours, extended the events service on AWS SQS, and pushed live part-state changes to the front end over WebSockets. A part could be red in SAP and green in ours, so idempotency and race conditions were the real work. Wrote the flows up as use cases and handed them to QA.

ReactJS Developer (Contractor)

Podium Pick'em Challenge. Remote. Apr 2024 - Jun 2024. Freelance, alongside John Deere.

- Responsive React front end with reusable components for a sports prediction platform tied to the 2024 Olympics, built to a tight deadline.

Ssr. .NET Developer

PwC. Remote. Aug 2022 - Sep 2023

- Co-led a three-person team modernizing legacy .NET Framework systems for US clients. Visiting US leadership recognized the team for the drop in open vulnerabilities that year.
- I was the person people came to on application security. Cleared 30+ Checkmarx findings and worked out staged .NET and NuGet upgrade paths for codebases full of vulnerable packages, without rewriting them. One package problem went all the way to Microsoft's internal team.
- Features, maintenance and WCAG 2.1 accessibility on PwC Policy on Demand, which is still running. Later, Excel reporting for tax calculation work.

.NET Developer

Ingenea S.R.L. Rosario, hybrid. Dec 2021 - Aug 2022

- .NET Core APIs and SQL Server stored procedures for clients in Argentina, Brazil and Mexico, Ternium among them, with data access split across repository and service layers.

EDUCATION

Universidad Abierta Interamericana

Bachelor's Degree in Technology Information Management (Licenciatura en Gestión de Tecnología Informática). Apr 2025 to expected Nov 2027, in progress.

Instituto Zona Oeste

Technical Degree in Software Development, a three-year technical program. Mar 2019 - Nov 2022. GPA 7.33.

LANGUAGES

Spanish, native. English, C2. German, B1 and working towards B2.